

Selection sort: The selection sort technique consists of entirely a selection phase in which the largest of the remaining elements is repeatedly placed in its proper position, i, at the end of the array. To do so the largest element is swapped with the element a[i].

```
void selection_sort(int a[], index)
{
    for(i=n-1;i>0;i--)
    {
        large=a[0];
        index=0;
        for(j=1;j<=i;j++)
        {
            if(a[j]>large)
            {
                large=a[j];
                index=j;
            }
        }
        a[index]=a[i];
        a[i]=large;
    }
}
```

Input Array

23, 67, 32, 14, 56, 45, 38

$n=7$

I Pass:

$i=6$

$large = a[0] = 23$

$index = 0$

if $67 > 23 \rightarrow large=67, index=1$

if $32 > 67$, false

if $14 > 67$, false

if $56 > 67$, false

if $45 > 67$, false

if $38 > 67$, false

At the of the I pass, $a[n-1] = a[6] = large=67$

$a[index] = a[1] = 38$

resultant Array, after I pass,

23, 38, 32, 14, 56, 45, 67

II Pass:

$i=5$

$large = a[0] = 23$

$index = 0$

if $38 > 23 \rightarrow \text{large}=38, \text{index}=1$
if $32 > 38$, false
if $14 > 38$, false
if $56 > 38 \rightarrow \text{large} = 56, \text{index} = 4$
if $45 > 56$, false

At the of the I pass, $a[n-2] = a[5] = \text{large}=56$
 $a[\text{index}] = a[4] = 45$
resultant Array, after II pass,

23, 38, 32, 14, 45, 56, 67

III Pass:

$i=4$

$\text{large} = a[0] = 23$

$\text{index} = 0$

if $38 > 23 \rightarrow \text{large}=38, \text{index}=1$
if $32 > 38$, false
if $14 > 38$, false
if $45 > 38 \rightarrow \text{large} = 45, \text{index} = 4$

At the of the I pass, $a[n-3] = a[4] = \text{large}=45$
 $a[\text{index}] = a[4] = 45$
resultant Array, after II pass,

23, 38, 32, 14, 45, 56, 67

IV Pass:

$i=3$

$large = a[0] = 23$

$index = 0$

if $38 > 23 \rightarrow large=38, index=1$

if $32 > 38$, false

if $14 > 38$, false

At the of the I pass, $a[3] = large=38$

$a[index] = a[1] = 14$

resultant Array, after IV pass,

23, 14, 32, 38, 45, 56, 67

V Pass:

$i=2$

$large = a[0] = 23$

$index = 0$

if $14 > 23$, false

if $32 > 23, \rightarrow large=32, index=2$

At the of the V pass, $a[2] = large=32$

$a[index] = a[2] = 32$

resultant Array, after V pass,

23, 14, 32, 38, 45, 56, 67

VI Pass:

i=1

large = a[0] = 23

index = 0

if 14>23, false

At the of the VI pass, a[1] = large=23

a[index] = a[0] = 14

resultant Array, after VI pass,

14, 23, 32, 38, 45, 56, 67